

Reinforcement Learning (RL) Algorithms in a Pick & Place (P&P) System for Autonomous Robotic Control

1. Project Description

In this project, we invite you to explore the application of RL algorithms to a P&P system. As we have seen in the earlier weeks, P&P is a fundamental robotic capability where an agent must locate, grasp and transport an object. It is used widely in industrial automation and logistics, with wide documentation in academic proceedings, and clear integration into creative robotics/media-arts spheres. Here, due to time constraints, we will focus on the *technical* aspect of this problem, leaving these integration pathways as a reflective task.

P&P can be formulated as four stages:

- **Perception and State Representation** - identifying and encoding relevant information about the robot, objects, and target locations within the environment.
- **Approach** - control the robotic manipulator to move its end-effector toward the object.
- **Grasp and Transport** - learning a stable grasp strategy and transporting the object while maintaining control and stability.
- **Placement** - accurately positioning and releasing the object at the desired target location.

Your task is to design, implement, and evaluate an RL-based control system that enables a simulated robotic arm to solve this task in a Unity-based environment. You are expected to define the environment, specify the observation and action spaces, and construct a reward function that meaningfully guides learning across the different phases of the task.

You are encouraged to use Unity in combination with the ML-Agents framework to implement and train your system. Robot models (such as Panda, UR5, or Kinova arms) may be imported via URDF files, and the simulation should be configured to support controlled experimentation, including randomisation of object positions and environmental conditions to evaluate generalisation performance. Training will be conducted using standard RL algorithms such as Proximal Policy Optimisation (PPO), as provided within the ML-Agents toolkit. Unity can then be used for scene design/management and visualisation purposes.

The project is structured to encourage an iterative development approach: beginning with simple reaching behaviours, progressing to object grasping, and culminating in a full pick-and-place pipeline. This staged progression is intended to support systematic debugging, evaluation, and refinement of both the learning environment and the agent's policy. Please refer to the P&P phases above.

This assessment will primarily evaluate your ability to formulate a robotics control problem as a reinforcement learning task - design appropriate observation, action, and reward structures, implement and train an RL agent within a simulation framework, and critically evaluate system performance across the different phases of the task. Successful submissions will demonstrate a clear understanding of reinforcement learning principles, and evidence systematic experimentation, analysis, and evaluation.

2. Objectives

The final project will test your ability to apply RL and robotic simulation concepts to a Pick and Place (P&P) system. In the project, you should demonstrate the following:

1. Understanding RL for robotic control, including how these approaches can be used to solve sequential decision-making problems in continuous control tasks such as robotic manipulation. Your report should reference relevant foundational work in RL and robotics where appropriate.
2. Formulation and implementation of an RL-based system. You are expected to define the problem in terms of observations, actions, and rewards, and clearly justify your design choices. This includes explaining how the robot state and environment are represented, how control signals are structured, and how the reward function guides learning across the different phases of the P&P task. If using frameworks such as Unity ML-Agents, you should explain why the chosen algorithms and tools are appropriate for the task, and annotate your code to demonstrate understanding of the implementation.

3. Simulation development and experimental evaluation - build and evaluate a working simulation of a robotic arm performing P&P. The environment should be clearly structured and reproducible, with appropriate use of robot models (e.g., imported URDF-based arms) and task randomisation where relevant. You are expected to evaluate system performance across different task stages, discussing success rates, learning stability, and failure cases. Visual documentation of training progress and agent behaviour should be included in your report (e.g., graphs of reward progression and example simulation outputs).

3. Report

A report showing the process of the project development and your independent inquiry is as important as the code files. It should clearly document your work. The report should include:

- **Abstract** - Summarise the key elements of your project, including its purpose. This section should provide a concise overview of what you have implemented
- **CCS Concepts** - Provide the ACM Computing Classification System (CCS) concepts that are relevant to your project. Use appropriate keywords and categories that reflect your project's focus on topics.
- **Keywords** - Include a list of keywords that represent the core topics of your project.
- **Introduction** - Provide an overview of the concept and objectives of your project. Explain how it fits within the broader context of RL in robotics.
- **Background** - Describe the research that informs your project. This could include a review of related works e.g relevant technologies, implementations or research papers that influenced your approach. Establish how your implementation relates to these inspirations.
- **Technical Implementation** - Detail the technical aspects of your project. This should include RL concepts (designing observation, actions and reward spaces) and how these have been implemented. An overview of the system architecture, including software modules and how they “talk” to one another, should also be included.
- **Results** - Present your process of *testing* and *evaluating* the system. Highlight what worked well, what challenges you encountered, and how you iterated upon technical issues.
- **Reflection** - Critically evaluate the outcomes of your project. Reflect on how effectively the project meets your original objectives. This section should also mention any limitations of your project and suggest areas for future development. This can include integration into your own creative practice - what systems could you design?

3. Submission

Due to time constraints, we will not make LaTeX formatting compulsory - although, if you are comfortable, we encourage you to use these tools. If this is the case, you can follow the [ACM Conference Proceedings Primary Article Template](#) and continue in Overleaf. If not, use a standard text editor/document formatting system, adhering to the report sections given above.

Project files should be organised into folders, so that after the zip extraction the structure is clear. Please use meaningful folder/file names and avoid a deep folder structure (nested).

LLM Disclaimer

Please ensure to clearly state and describe how you have used the AI tools for your project. Like any other sources you may consider using in assessments such as a journal, book or database, you must properly reference (cite) AI in your work. The [Cite Them Right](#) website provides guidance on how to reference generative AI in work. You can also refer to [UAL's Student Guide to Generative AI](#).